

for the associated risk. An asset with a negative Sharpe ratio is punishing the investor with negative returns and volatility.

Importantly, absolute returns are only half the story for the Sharpe ratio. An asset with lower absolute returns can have a higher Sharpe ratio than a high-flying asset that experiences extreme volatility. For example, consider an equity asset that has an expected return of 12 percent with a volatility of 10 percent, versus a bond with an expected return of 5 percent but volatility of 3 percent. The former has a Sharpe ratio of 1.2 while the latter of 1.67 (assuming a risk-free rate of 0 percent). The ratio provides a mathematical method to compare how different assets compensate the investor for the risk taken, making bonds and equities, or apples and oranges, more comparable.

Correlation of Returns and the Efficient Frontier

One of the key breakthroughs of modern portfolio theory was to show that a riskier asset can be added to a portfolio, and if its behavior differs significantly from the preexisting assets in that portfolio, it can actually decrease the overall risk of the portfolio. How can a risky asset make a portfolio less risky? The key is *correlation of returns*.

Correlation simply measures how assets move in relation to one another. The measurement ranges from a value of +1 to -1. If assets are perfectly positively correlated, then they move in tandem: if one is up 10 percent, the other is up 10 percent as well, for a score of +1. Similarly, if they are perfectly negatively correlated at -1, then when one is up 10 percent the